Assignment 4

CS 6117 (Fall 2025)

In this exercise, we will work through a common application of gluing to prove *canonicity*: every closed term of a given type has a particular form.

Exercise 1. Let $\mathbf{C}$ be a (locally small) cartesian closed category.

1. We will be gluing along the functor $\mathbf{C}(1,-) : \mathbf{C} \to \mathbf{Set}$, also known as the *global sections* functor. What are the objects and morphisms of the comma category $Id_{\mathbf{Set}} \downarrow \mathbf{C}(1,-)$? We will call this glued category $\mathbf{Gl}(\mathbf{C})$.

2. Show that $\mathbf{C}(1,-)$ preserves binary products and that $\mathbf{Set}$ has binary pullbacks.

By the gluing theorem (CfT 4.10.3), the category $\mathbf{Gl}(\mathbf{C})$ is cartesian closed and the forgetful functor $U : \mathbf{Gl}(\mathbf{C}) \to \mathbf{C}$ is a cartesian closed functor (i.e., it preserves products and exponentials). The glued category is also known as a *scone* or the *Freyd cover*.

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1. The comma category $\mathrm{Id}_{\mathbf{Set}} \downarrow \mathrm{C}(1,-)$ consists of Objects: Triples (S, A, f), where $S \in \mathbf{Ob}(\mathbf{Set})$, $A \in \mathbf{Ob}(\mathbf{C})$, and $f : S \to \mathbf{C}(1, A)$ is a function in $\mathbf{Set}$ since $\mathbf{C}$ is locally small, $\mathbf{C}(1, A)$ is a set. Morphisms: From (S, A, f) to (S', A', f') is a pair (h, k), where $h : S \to S'$ in $\mathbf{Set}$ and $k : A \to A'$ in $\mathbf{C}$, such that the following square commutes:

$$\begin{array}{ccc} S & \xrightarrow{f} & \mathbf{C}(1, A) \\ {\scriptstyle h}\downarrow & & \downarrow{\scriptstyle \mathbf{C}(1,k)} \\ S' & \xrightarrow{f'} & \mathbf{C}(1, A') \end{array}$$

   $\mathbf{C}(1, k) \circ f = f' \circ h$, where $\mathbf{C}(1, k)$ post-composes with k.

2. Let $A, B \in \mathbf{Ob}(\mathbf{C})$. Since $\mathbf{C}$ is cartesian closed, hence has finite products, and 1 is terminal, $\mathbf{C}(1, A \times B) \cong \mathbf{C}(1, A) \times \mathbf{C}(1, B)$, naturally in A and B, by the universal property of products: a morphism $1 \to A \times B$ is equivalent to a pair $(1 \to A, 1 \to B)$. $\mathbf{Set}$ has binary pullbacks. Given functions $f : X \to Z$ and $g : Y \to Z$, the pullback is $X \times_Z Y = \{(x, y) \in X \times Y \mid$

$f(x) = g(y)\}$, with projections $\pi_X : (x, y) \mapsto x$ and $\pi_Y : (x, y) \mapsto y$. This satisfies the universal property: for any $h : W \to X$, $k : W \to Y$ with $f \circ h = g \circ k$, the unique mediating map is $w \mapsto (h(w), k(w))$. By the gluing theorem (CfT 4.10.3), $\mathrm{Gl}(\mathbf{C})$ is cartesian closed, and $U : \mathbf{Gl}(\mathbf{C}) \to \mathbf{C}$ (the forgetful functor $(S, A, f) \mapsto A$) preserves products and exponentials.

Exercise 2. Let $Th = (Sg, Ax)$ be a $\lambda\times$ theory with a single base type nat, and function symbols Z, S, and Rec_τ with the following typing rules:

$$\textsc{Zero}\ \frac{}{\Gamma \vdash \mathsf{Z} : \mathsf{nat}} \qquad \textsc{Succ}\ \frac{\Gamma \vdash t : \mathsf{nat}}{\Gamma \vdash \mathsf{S}\, t : \mathsf{nat}} \qquad \textsc{Rec}\ \frac{\Gamma \vdash t_0 : \tau \qquad \Gamma \vdash t_S : \tau \to \tau}{\Gamma \vdash \mathsf{Rec}_\tau\, t_0\, t_S : \mathsf{nat} \to \tau}$$

Along with the usual equations for $\lambda\times$ theories, we take the following axioms:

$$\beta\text{-}\textsc{Zero}\ \frac{\Gamma \vdash t_0 : \tau \qquad \Gamma \vdash t_S : \tau \to \tau}{\Gamma \vdash (\mathsf{Rec}_\tau\, t_0\, t_S\, \mathsf{Z}) = t_0 : \tau}$$

$$\beta\text{-}\textsc{Succ}\ \frac{\Gamma \vdash t_0 : \tau \qquad \Gamma \vdash t_S : \tau \to \tau \qquad \Gamma \vdash t : \mathsf{nat}}{\Gamma \vdash \mathsf{Rec}_\tau\, t_0\, t_S\, (\mathsf{S}\, t) = t_S\, (\mathsf{Rec}_\tau\, t_0\, t_S\, t) : \tau}$$

$$\eta\text{-}\textsc{Rec}\ \frac{\begin{array}{c}\Gamma \vdash t_0 : \tau \qquad \Gamma \vdash t_S : \tau \to \tau \qquad \Gamma \vdash f : \mathsf{nat} \to \tau \\ \Gamma \vdash f\, \mathsf{Z} = t_0 : \tau \qquad \Gamma, x : \mathsf{nat} \vdash f\, (\mathsf{S}\, x) = t_S\, (f\, x)\end{array}}{\Gamma \vdash f = \mathsf{Rec}_\tau\, t_0\, t_S}$$

We will assume that there are no other function symbols, base types, or axioms.

The categorical version of natural numbers are called *natural number objects* (nno). Formally, a nno consists of a triple $(N, Z : 1 \to N, S : N \to N)$ such that for every triple $(A, x_0 : 1 \to A, x_S : A \to A)$, there exists a unique arrow $u : N \to A$ such that $u \circ Z = x_0$ and $u \circ S = x_S \circ u$. If a category has a nno, then it is unique up to isomorphism. (See nlab for more details.)

We write $\mathbf{Cl}$ for the category where objects are types $[\tau]$, and morphisms $[\tau_1] \to [\tau_2]$ are represented by programs/typing judgments $[x : \tau_1 \vdash t : \tau_2]$ modulo the equational theory, i.e., if $x : \tau_1 \vdash t = u : \tau_2$ then $[x : \tau_1 \vdash t : \tau_2]$ and $[x : \tau_1 \vdash u : \tau_2]$ are the same morphism. The identity morphism is represented by the program $[x : \tau \vdash x : \tau]$, and composition is given by substitution. The $\mathbf{Cl}$ has a nno given by $([\mathsf{nat}], [\cdot \vdash \mathsf{Z} : \mathsf{nat}], [x : \mathsf{nat} \vdash \mathsf{S}\, x : \mathsf{nat}])$.

You may assume that $\mathbf{Cl}$ is the classifying category of the theory Th, so it satisfies the following universal property: for every model M of Th in cartesian closed category $\mathbf{D}$ with a nno, there is exactly one functor $\mathbf{Cl} \to \mathbf{D}$ preserving the nno and ccc structure and mapping the generic model to M. In fact, every nno determines a model of our theory, and vice versa.

1. Given the equations above, what is $\mathsf{Rec}_\tau\, t_0\, t_S\, (\mathsf{S}\, \mathsf{S}\, \mathsf{Z})$ equal to?

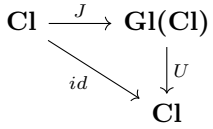
2. This type theory can be interpreted in **Set** by setting $[\![\mathsf{nat}]\!] = \mathbb{N}$, where we write $\mathbb{N}$ for the set of natural numbers (from regular mathematics). How are the constants Z, S, and Rec_τ interpreted?

3. Assume that $\mathbf{C}$ has a natural number object $(N, Z : 1 \to N, S : N \to N)$. Show that (X, i, N) is a natural number object in $\mathbf{Gl}(\mathbf{C})$, where:

$$X \triangleq \{f \in \mathbf{C}(1, N) \mid \exists n \in \mathbb{N}. f = S^n \circ Z\},$$

   S^n is S composed with itself n times, and $i : X \to \mathbf{C}(1, N)$ is set inclusion.

4. Since the forgetful functor $U : \mathbf{Gl}(\mathbf{Cl}) \to \mathbf{Cl}$ maps the nno from the previous part to $[\mathsf{nat}] : \mathbf{Cl}$, it preserves the nno. Show that there is a functor $J : \mathbf{Cl} \to \mathbf{Gl}(\mathbf{Cl})$ making the following diagram commute:



   [Hint: your answer can be quite short—the heavy lifting is already done by the universal property of $\mathbf{Cl}$.]

5. Conclude if $\cdot \vdash t : \mathsf{nat}$ then there is a natural number $n \in \mathbb{N}$ such that

$$\cdot \vdash t = \underbrace{\mathsf{S} \cdots \mathsf{S}}_{n}\ \mathsf{Z} : \mathsf{nat}.$$

   [Hint: where does the morphism $[\cdot \vdash t : \mathsf{nat}]$ in $\mathbf{Cl}$ go?]

6. Adding function symbols can break canonicity. For instance, if we add a new function symbol $\vdash \mathsf{Foo} : \mathsf{nat}$ and no new axioms, then Foo is not equal to any natural number, so canonicity should fail. What breaks in the proof? [Hint: is $[\mathsf{nat}]$ still a nno?]

   In contrast, adding new equations cannot break canonicity, even though it may lead to trivial models. For instance, if we added the equation $\mathsf{Z} = \mathsf{S}\ \mathsf{Z}$, the above argument would still technically go through. However, any model of the theory would be degenerate (e.g., all programs of type nat would be equal).

We were able to automatically find a functor $J : \mathbf{Cl} \to \mathbf{Gl}(\mathbf{Cl})$ once we showed that the glued category was ccc and had a nno, since all the hard work is in showing the universal property of $\mathbf{Cl}$. If we add other extensions to the language—especially extensions that aren't characterized by universal properties like ccc and nno—then carrying out the proof requires defining the functor $J : \mathbf{Cl} \to \mathbf{Gl}(\mathbf{Cl})$ explicitly.

Defining this functor is the categorical version of proving a property syntactically by logical relations. In particular, J maps objects of $\mathbf{Cl}$ to objects of the

glued category; picking this mapping is a categorical version of defining a logical relation. Likewise, J maps morphisms in **Cl** to morphisms in **Gl**(**Cl**); defining this mapping and showing that the target are indeed morphisms in the glued category are the key things to check in a standard proof by logical relations.

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1. $\vdash \mathrm{Rec}_\tau t_0 t_S(\mathrm{SSZ}) = t_S(\mathrm{Rec}_\tau t_0 t_S(\mathrm{SZ})) = t_S(t_S(\mathrm{Rec}_\tau t_0 t_S \mathrm{Z})) = t_S(t_S t_0) : \tau$

$$\begin{aligned}\mathrm{Rec}_\tau t_0 t_S(SSZ) &= t_S(\mathrm{Rec}_\tau t_0 t_S(SZ)) \quad (\text{by } \beta\text{-Succ with } t = SZ)\\ &= t_S(t_S(\mathrm{Rec}_\tau t_0 t_S Z)) \quad (\text{by } \beta\text{-Succ with } t = Z)\\ &= t_S(t_S t_0) \quad (\text{by } \beta\text{-Zero}).\end{aligned}$$

2. Interpret $[\![\mathrm{nat}]\!] = \mathbb{N} = \{0, 1, 2, \dots\}$. Then, $[\![\mathrm{Z}]\!] = 0 \in \mathbb{N}$, $[\![\mathrm{S}]\!](n) = n + 1$, $[\![\mathrm{Rec}_\tau]\!](t_0, t_S)(n) = f(n)$, where $f : \mathbb{N} \to [\![\tau]\!]$ is defined recursively by $f(0) = t_0$ and $f(m + 1) = t_S(f(m))$. This satisfies the axioms: β-Zero gives $f(0) = t_0$, β-Succ gives $f(n + 1) = t_S(f(n))$, and η-Rec ensures uniqueness.

3. Assume C is a category with a natural number object (NNO) $(N, Z : 1 \to N, S : N \to N)$. The glued category $\mathrm{Gl}(C)$ is the comma category $\mathbf{Set} \downarrow C(1, -)$, where:

   Objects are triples (Y, q, B) with $Y \in \mathbf{Set}$, $B \in C$, $q : Y \to C(1, B)$ Morphisms $(f : Y \to Y', g : B \to B')$ satisfy $C(1, g) \circ q = q' \circ f$.

   The forgetful functor $U : \mathrm{Gl}(C) \to C$ sends $(Y, q, B) \mapsto B$. We show that (X, i, N) is an NNO in $\mathrm{Gl}(C)$, where $X = f \in C(1, N) \mid \exists n \in \mathbb{N}.\ f = S^n \circ Z$ and $i : X \to C(1, N)$ is the inclusion. The zero morphism is the pair $(\hat{Z} : 1 \to X, Z : 1 \to N)$, where $\hat{Z}() = Z$, and it commutes since $i(\hat{Z}()) = Z$. The successor is the pair $(\hat{S} : X \to X, S : N \to N)$, where $\hat{S}(f) = S \circ f$, and commutes since $i(\hat{S}(f)) = S \circ f = S \circ i(f)$.

   To verify the universal property: for any object $(Y, q, A) \in \mathrm{Gl}(C)$, zero morphism $(u_Y : 1 \to Y, u : 1 \to A)$ with $q \circ u_Y = C(1, u)$, and successor $(v_Y : Y \to Y, v : A \to A)$ with $q \circ v_Y = C(1, v) \circ q$, there is a unique recursor pair $(r : X \to Y, \mathrm{rec} : N \to A)$ commuting $q \circ r = C(1, \mathrm{rec}) \circ i$. Since (N, Z, S) is NNO in C, there is unique $\mathrm{rec} : N \to A$ with $\mathrm{rec} \circ Z = u$ and $\mathrm{rec} \circ S = v \circ \mathrm{rec}$. Define $r : X \to Y$ by $r(S^n \circ Z) = v_Y^n(u_Y(*))$. Induction shows $q \circ r(S^n \circ Z) = v^n(u) = \mathrm{rec} \circ S^n \circ Z = \mathrm{rec} \circ i(S^n \circ Z)$. Uniqueness follows from the inductive definition $r(\hat{Z}) = u_Y(*)$, $r \circ \hat{S} = v_Y \circ r$, mirroring the NNO property on the set level.

4. The forgetful $U : \mathrm{Gl}(C) \to C$ preserves the NNO since $U(X, i, N) = N$ and maps structure accordingly. The functor $J : C \to \mathrm{Gl}(C)$ is defined by $J(B) = (C(1, B), \mathrm{id}, B)$ and $J(g : B \to B') = (C(1, g), g)$. Then $U \circ J = \mathrm{id}_C$, making the diagram commute

5. By the universal property of the classifying category Cl (the free CCC with NNO generated by the theory), every term t with $\vdash t : \mathrm{nat}$ is interpreted in

a bit confused here, could use more details. for instance how do we use completeness?

Gl(Set) as an element of X, the standard numerals. Since interpretations preserve equalities and the structure maps t to a unique numeral in models, and by completeness, $\vdash t = S^n Z$ for some n (where n is the value in the **Set** model).

6. Adding `Foo` : nat without axioms allows `Foo` as a non-standard global element not equal to any $S^n Z$. The construction (X, i, N) no longer satisfies the universal property of an NNO in $\mathrm{Gl}(C)$, as recursors can distinguish `Foo` from numerals, breaking uniqueness. Adding equations like $Z = SZ$ degenerates the theory, making all numerals equal, again violating the NNO property.

Exercise 3. Adjoint functors are everywhere. In the following exercise, you will work out a few examples.

1. Let $(C, \leq_C)$ and $(D, \leq_D)$ be two partially ordered sets. Posets can be viewed as posetal categories, i.e., posets where there is at most one morphism between any two objects. Spell out what it means for $F : \mathbf{C} \to \mathbf{D}$ and $G : \mathbf{D} \to \mathbf{C}$ to be adjoint functors. Pairs of functions like F and G are called a *Galois connection*, and play a central role in the theory of abstract interpretation.

2. Let $\Delta : \mathbf{Set} \to \mathbf{Set} \times \mathbf{Set}$ be the *diagonal* functor, which maps objects A to (A, A) (and morphisms to?). Describe the left and right adjoints of Δ.

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1. Given two posets $(C, \leq_C)$ and $(D, \leq_D)$, we can view them as categories: Objects are the elements of the poset. There is a unique morphism $c \to c'$ iff $c \leq_C c'$ (and no morphism otherwise). Composition and identities follow from transitivity and reflexivity of $\leq_C$. Functors between such categories are precisely monotone (order-preserving) functions. A functor $F : C \to D$ satisfies $c \leq_C c' \implies F(c) \leq_D F(c')$. For $F : C \to D$ and $G : D \to C$ to form an adjunction $F \dashv G$ (with F left adjoint to G), the general definition requires a natural bijection of hom-sets: $Hom_D(F(c), d) \cong Hom_C(c, G(d))$ for all objects $c \in C$, $d \in D$, natural in c and d. In posetal categories, hom-sets are either empty or singletons: $Hom_D(F(c), d)$ has one element if $F(c) \leq_D d$, and is empty otherwise. The same holds for $\hom_C(c, G(d))$. Thus, the bijection reduces to an equivalence of truth values, and the hom-sets are isomorphic iff both are empty or both are singletons. This yields $F(c) \leq_D d \iff c \leq_C G(d)$ for all $c \in C$, $d \in D$. F determines the least upper bound relative to G, $F(c)$ is the smallest element in D such that $c \leq_C G(F(c))$. Similarly, $G(d)$ is the largest in C such that $F(G(d)) \leq_D d$.

2. The diagonal $\Delta : \mathbf{Set} \to \mathbf{Set} \times \mathbf{Set}$ sends a set A to (A, A) and a function $f : A \to B$ to $(f, f) : (A, A) \to (B, B)$.

   Left adjoint (coproduct): $+ : \mathbf{Set} \times \mathbf{Set} \to \mathbf{Set}$ with $(A, B) \mapsto A + B = (A \times \{0\}) \cup (B \times \{1\})$. The bijection is $\mathbf{Set}(A + B, C) \cong (\mathbf{Set} \times \mathbf{Set})((A, B), \Delta(C)) = \mathbf{Set}(A, C) \times \mathbf{Set}(B, C)$, via pairs of maps inducing a unique map on the coproduct

   Right adjoint (product): $\times : \mathbf{Set} \times \mathbf{Set} \to \mathbf{Set}$ with $(A, B) \mapsto A \times B$. The bijection is $\mathbf{Set}(C, A \times B) \cong (\mathbf{Set} \times \mathbf{Set})(\Delta(C), (A, B)) = \mathbf{Set}(C, A) \times \mathbf{Set}(C, B)$, via maps to the product induced by pairs of maps.

Exercise 4 (Category of the Week). Given functors $F : \mathbf{C} \to \mathbf{D}$ and $G : \mathbf{E} \to \mathbf{D}$, the *comma category* $F \downarrow G$ has as objects triples $(A \in \mathbf{Ob}(\mathbf{C}), B \in \mathbf{Ob}(\mathbf{E}), f : F(A) \to G(B))$, sometimes written $F(A) \xrightarrow{f} G(B)$ for short. A morphism from (A, B, f) to (X, Y, g) is given by maps $p : A \to X, q : B \to Y$ forming a commuting square:

$$\begin{array}{ccc} F(A) & \xrightarrow{f} & G(B) \\ \downarrow{\scriptstyle F(p)} & & \downarrow{\scriptstyle G(q)} \\ F(X) & \xrightarrow{g} & G(Y) \end{array}$$

1. Show that if $\mathbf{C}$, $\mathbf{D}$ and $\mathbf{E}$ all have products and $G : \mathbf{C} \to \mathbf{E}$ preserves products, then $F \downarrow G$ also has products. [Hint: Lemma 4.10.3 in CfT sketches what products look like; you should spell out the details. That lemma also describes exponentials—which you don't have to check—but the right-side vertical morphism is incorrect since it assumes Γ preserves exponentials. The errata corrects this figure.]

The comma category looks a bit strange but it generalizes many examples of categories, so properties of the comma category carry over to other categories. For each of the following examples, briefly describe the category $F \downarrow G$. We have seen many of these categories before.

2. $\mathbf{C}$ and $\mathbf{D}$ are both the singleton/terminal category.
3. $\mathbf{E}$ is the singleton category.
4. $\mathbf{C}$ is the singleton category, $\mathbf{D} = \mathbf{E}$, and G is the identity functor.
5. $\mathbf{C} = \mathbf{D} = \mathbf{E}$, and F and G are the identity functors.

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1. Assume $\mathbf{C}$, $\mathbf{D}$, and $\mathbf{E}$ have (finite) products, and G preserves products (i.e., $G(B_1 \times B_2) \cong G(B_1) \times G(B_2)$ via canonical isomorphisms, and similarly for projections). Let $u = (A_1 \in \mathbf{C}, B_1 \in \mathbf{E}, f_1 : F(A_1) \to G(B_1))$ and $v = (A_2, B_2, f_2 : F(A_2) \to G(B_2))$ be objects in $F \downarrow G$. The candidate product is $w = (A_1 \times_{\mathbf{C}} A_2, B_1 \times_{\mathbf{E}} B_2, f : F(A_1 \times A_2) \to G(B_1 \times B_2))$. Let $A = A_1 \times_{\mathbf{C}} A_2$ with projections $\pi_1^A : A \to A_1$, $\pi_2^A : A \to A_2$. Let $B = B_1 \times_{\mathbf{E}} B_2$ with projections $\pi_1^B : B \to B_1$, $\pi_2^B : B \to B_2$. Since G preserves products, $G(B) \cong G(B_1) \times_{\mathbf{D}} G(B_2)$ with projections $G(\pi_1^B), G(\pi_2^B)$. Define f as the unique morphism in $\mathbf{D}$ by universality of the product $G(B_1) \times G(B_2)$, satisfying $G(\pi_1^B) \circ f = f_1 \circ F(\pi_1^A), \quad G(\pi_2^B) \circ f = f_2 \circ F(\pi_2^A)$. Projections in $F \downarrow G$, $\pi_u = (\pi_1^A, \pi_1^B) : w \to u$. $G(\pi_1^B) \circ f = f_1 \circ F(\pi_1^A)$ by definition. Similarly for $\pi_v = (\pi_2^A, \pi_2^B) : w \to v$. Let $z = (S \in \mathbf{C}, T \in \mathbf{E}, h : F(S) \to G(T))$ with morphisms $\phi_u = (p_1 : S \to A_1, q_1 : T \to B_1) : z \to u$ and $\phi_v = (p_2 : S \to A_2, q_2 : T \to B_2) : z \to v$. The mediator is $\phi = (\langle p_1, p_2 \rangle_{\mathbf{C}} : S \to A, \langle q_1, q_2 \rangle_{\mathbf{E}} : T \to B)$. Commutativity $G(\langle q_1, q_2 \rangle) \circ h = f \circ F(\langle p_1, p_2 \rangle)$ holds by the given commutativities of ϕ_u, ϕ_v, the definition of f, and G's preservation ensuring $G(\langle q_1, q_2 \rangle) = \langle G(q_1), G(q_2) \rangle$. Uniqueness follows from product universality in $\mathbf{C}$ and $\mathbf{E}$.

2. If $\mathbf{C}$ and $\mathbf{D}$ are both the singleton (terminal) category $\mathbf{1}$ (one object $*$, one morphism id_*), then F, G are unique. Assuming $\mathbf{E} = \mathbf{1}$ too (to make sense), $F \downarrow G$ has one object $(*, *, \mathrm{id}_*)$ and one morphism, so it is the singleton/terminal category.

why doesn't this problem make sense with general E?

3. If $\mathbf{E} = \mathbf{1}$ (singleton), then $G : \mathbf{1} \to \mathbf{D}$ picks a fixed object $K = G(*)$ in $\mathbf{D}$. Objects of $F \downarrow G$ are $(A, *, f : F(A) \to K)$; morphisms $(p : A \to X, \mathrm{id}_*)$ with $f = g \circ F(p)$. This is the slice category $F \downarrow K$, generalizing objects under K via F.

4. If $\mathbf{C} = \mathbf{1}$ (singleton), $\mathbf{D} = \mathbf{E}$, and $G = \mathrm{id}_{\mathbf{D}}$, then $F : \mathbf{1} \to \mathbf{D}$ picks $L = F(*)$. Objects are $(*, B, f : L \to B)$; morphisms $(\mathrm{id}_*, q : B \to Y)$ with $q \circ f = g$. This is the slice category $L \downarrow \mathbf{D}$ (objects are morphisms from fixed L).

5. If $\mathbf{C} = \mathbf{D} = \mathbf{E}$ and $F = G = \mathrm{id}_{\mathbf{C}}$, then objects are $(A, B, f : A \to B)$; morphisms $(p : A \to X, q : B \to Y)$ with $q \circ f = g \circ p$. This is the arrow category $\mathbf{C}^{\to}$, where objects are morphisms of $\mathbf{C}$ and morphisms are commuting squares.

Optional Exercises. These exercises explore important adjoints in our favorite category: slice categories $\mathbf{Set}/A$.

1. Let $f : A \to B$ be a morphism in $\mathbf{Set}$. Then, there is a *base change* functor $f^* : \mathbf{Set}/B \to \mathbf{Set}/A$. Note the change in direction: in A1 you saw a different functor. On objects, this functor pulls back $g : X \to B$ along $f : A \to B$. (Recall that in the first exercise, you showed that $\mathbf{Set}$

has pullbacks.) Describe how f^* acts on morphisms in $\mathbf{Set}/B$, and show that it is a functor.

2. There is a right adjoint to $f^* : \mathbf{Set}/B \to \mathbf{Set}/A$, usually called $\Pi_f : \mathbf{Set}/A \to \mathbf{Set}/B$. Describe what this functor does. In models of dependent type theory, Π_f is used to form the *dependent product* type.

3. There is also a left adjoint to $f^* : \mathbf{Set}/B \to \mathbf{Set}/A$, usually called $\Sigma_f : \mathbf{Set}/A \to \mathbf{Set}/B$. Describe what this functor does. In models of dependent type theory, Σ_f is used to form the *dependent sum* type.

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1. On objects: For an object $g : X \to B$ in $\mathbf{Set}/B$, $f^*(g)$ is the first projection $p_1 : P \to A$ from the pullback P of g along f: $P = \{(a, x) \in A \times X \mid f(a) = g(x)\}$, $p_1(a, x) = a$. The other projection is $p_2 : P \to X$, $(a, x) \mapsto x$, and the square $f \circ p_1 = g \circ p_2$ commutes by construction.) On morphisms: For a morphism $h : (X, g) \to (Y, k)$ in $\mathbf{Set}/B$ (i.e., $h : X \to Y$ with $k \circ h = g$), $f^*(h)$ is the induced map on pullbacks: $f^*(h) : P \to Q$, $(a, x) \mapsto (a, h(x))$, where $Q = \{(a, y) \in A \times Y \mid f(a) = k(y)\}$ is the pullback for k. This preserves the structure: $q_1(f^*(h)(a, x)) = a = p_1(a, x)$ and $q_2(f^*(h)(a, x)) = h(x) = h \circ p_2(a, x)$, and since $f(a) = g(x) = k(h(x))$, the image is in Q. Then for functor, Identity: $f^*(\mathrm{id}_X)(a, x) = (a, \mathrm{id}_X(x)) = (a, x) = \mathrm{id}_P$. Composition: For $h' : (Y, k) \to (Z, l)$, $f^*(h' \circ h)(a, x) = (a, h'(h(x))) = f^*(h') \circ f^*(h)(a, x)$.

2. The functor Π_f (dependent product along f) is the right adjoint to f^*, satisfying $Hom_{\mathbf{Set}/A}(f^*(g), m) \cong Hom_{\mathbf{Set}/B}(g, \Pi_f(m))$ naturally. On objects: For $m : Z \to A$ in $\mathbf{Set}/A$, think of fibers $Z_a = m^{-1}(a)$), $\Pi_f(m) : W \to B$ where $W = \left\{(b, s) | b \in B,\ s : f^{-1}(b) \to Z,\ \forall a \in f^{-1}(b),\ m(s(a)) = a\right\}$ with projection $\pi : (b, s) \mapsto b$. On morphisms: For a morphism $n : (Z, m) \to (Z', m')$ in $\mathbf{Set}/A$, $\Pi_f(n) : W \to W'$ sends $(b, s) \mapsto (b, n \circ s)$, preserving fibers since $m'(n(s(a))) = m(s(a)) = a$. Adjoint follows from the universal property of pullbacks and function spaces in Set. In dependent type theory, Π_f interprets dependent function types $\Pi_{a:A} Z_a$, where the domain depends on f(a)

3. The functor Σ_f (dependent sum along f) is the left adjoint to f^*, with $Hom_{\mathbf{Set}/B}(\Sigma_f(m), g) \cong Hom_{\mathbf{Set}/A}(m, f^*(g))$ naturally. On objects: For $m : Z \to A$ in $\mathbf{Set}/A$, $\Sigma_f(m) = f \circ m : Z \to B$. On morphisms: For $n : (Z, m) \to (Z', m')$, $\Sigma_f(n) = n : (Z, f \circ m) \to (Z', f \circ m')$ (commutes since $f \circ m' \circ n = f \circ m$). Adjoint holds by composition's universality in Set. In type theory, Σ_f models dependent sum types $\Sigma_{a:A} Z_a$, projecting to B via f.
